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Time of concentration: a paradox in modern hydrology

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Abstract The time of concentration is a primary parameter for a variety of modern hydrological models adopted in professional and scientific communities. Nevertheless, a universally accepted working definition of this parameter is currently lacking and several definitions can be found in the technical literature along with related estimation procedures. This study brings to light the inherent variability of these definitions through the empirical analysis of four small basins. These case studies demonstrate that available approaches for the estimation of the time of concentration may yield numerical predictions that differ from each other by up to 500%.

Key words time of concentration; travel time; lag time; ungauged basins

Temps de concentration: un paradoxe dans l'hydrologie moderne

Résumé Le temps de concentration est un important paramètre de nombreux modèles hydrologiques modernes utilisés par les professionnels et les scientifiques. Il n'existe cependant pas de définition universellement acceptée de ce paramètre, et on peut trouver dans la littérature plusieurs définitions ainsi que des procédures d'estimation connexes. Cette étude met en lumière la variabilité inhérente de ces définitions à travers l'analyse empirique de quatre petits bassins. Ces études de cas démontrent que les approches disponibles pour l'estimation des temps de concentration peuvent produire des estimations numériques qui diffèrent les unes des autres jusqu'à 500%.

Mots clefs temps de concentration; temps de déplacement; temps de latence; bassins non jaugés

INTRODUCTION

Hydrograph design for assigned return periods is a crucial aspect of hydrological analyses for its practical implications. Government institutions' guidelines (that is, in Italy: PAI 2006; in the US: FEMA 2010) and traditionally used handbooks of hydrology (see for example Chow *et al.* 1988, Seybert 2006) report several hydrograph design procedures that are based on the time of concentration (T_c). This parameter is especially relevant for ungauged basins and continues

to find application in recent hydrological models (USACE 2001, SWAT: Neitsch *et al.* 2005, Aronica and Candela 2007, Koutroulis and Tsanis 2010). In spite of the diffuse use of the time of concentration, a unique working definition is currently not available. However, a variety of “computational” and “theoretical” definitions is methodologically proposed and practically adopted in the literature.

Indeed, when both rainfall and runoff observations are available, six computational definitions for the time of concentration are reported in McCuen

(2009) and references therein: (i) the time from the end of rainfall excess to the inflection point on the total storm hydrograph; (ii) the time from the centre of mass of rainfall excess to the centre of mass of direct runoff; (iii) the time from the maximum rainfall intensity to the time of the peak discharge; (iv) the time from the centre of mass of rainfall excess to the time of the peak of direct runoff; (v) the time from the centre of mass of rainfall excess to the time of the peak of total runoff; and (vi) the time from the start of the total runoff to the time of the peak discharge of the total runoff. Note that definition (ii) coincides with the traditional concept of “lag time”. However, this parameter reflects the storage properties of watersheds, that are fundamentally different from the time of concentration.

In addition to the above six computational definitions, the following theoretical definitions are also used in the technical literature (McCuen 2009): (a) the time that a drop of rainwater spends to arrive to the basin outlet section starting from the most hydraulically distant point of the basin and (b) the time from the end of rainfall excess to the time of the end of direct runoff.

The simultaneous use of these eight different definitions for the time of concentration, along with the sparseness of available direct measurements, sets a further practical challenge for establishing a unique and unbiased estimation procedure for this parameter. To the best of the authors' knowledge, direct observations are restricted to the work of Holecek and Vocel (1965), Calkins and Dunne (1970), and Pilgrim (1975), where radioactive and chemical tracers are employed. Unfortunately, coarse space and time resolutions of such data have limited their potential of shedding light on the physical meaning of the time of concentration.

In practice, T_c is generally estimated by using empirical formulas at the basin scale. Giandotti's formula (Giandotti 1934) is extensively used in Italy, while the Kirpich (Kirpich 1940) and NRCS (National Research Conservation Service; Mockus 1961 unpublished report, Folmar *et al.* 2007) formulas are widely adopted in the USA. Although these formulas are very well accepted in the applied hydrology community, information on their technical foundations is limited. While it is known that they are based on the computational definition (i), the nature and number of observed flood events used in their calibration are not fully understood. The selection of rainfall–runoff events for calibrating these formulas is very important since T_c depends on the peak

flow. Indeed, the linear hypothesis that allows for considering T_c to be quasi-invariant with respect to the rainfall intensity can only be approximately verified for flood events with a high return period (Dooge 1973). Consequently, empirical approaches generally overestimate T_c when all the available flood events are considered in the calibration phase, irrespective of their magnitude.

In addition to these three widely used formulas and similar empirical relations available in the technical literature (see for example Wong 2005), an alternative approach, with conceptual and practical promise, is proposed by NRCS (1997). This strategy consists of estimating the flow velocity in the drainage network according to the previously mentioned theoretical definition (i). This approach, although more physically-based, is affected by several uncertainties. These are mainly attributed to the selection and implementation of the flow velocity formulas as well as to the drainage path identification. Results obtained with this methodology are also not very consistent (Fang *et al.* 2007, 2008, Pavlovic and Moglen 2008, McCuen 2009).

In this paper, selected definitions and related estimation procedures are specialized to the analysis of four case studies with the two-fold aim of drawing a critical comparison among available methodologies and providing a practical insight of a paradox of modern hydrology, that is, the definition and the estimation of T_c .

The specific objectives of this paper are:

- to elucidate the considerable variability of T_c estimations stemming from computational definition implemented on observed data;
- to ascertain the inherent limitations of empirical formulas in estimating T_c ; and
- to comment on the excessively large number of hypotheses required in the NRCS approach, that, in turn, determines an unexpected variability in the estimation of T_c .

The four case studies are presented in the next section, the adopted methods are then introduced in the following section, and then results and comments are reported.

CASE STUDIES DESCRIPTION

Four small USA basins are selected to benchmark the available definitions and understand their limitations, similarities and differences. These drainage areas are limited and their extent varies in the range

13–120 km². This allows us to retain the so-called linear hypothesis and adopt T_c models for small basin hydrological analysis.

The selected watersheds are:

- (1) Brazos basin, Cow Bayou Sub. 4 near Bruceville (lat. 31°19'59", long. 97°16'02", code 08096800); the drainage area is 13.08 km²; elevations vary from 180 to 266 m, with a mean slope of 5.9%; 47 rainfall–runoff observed events from 1959–1975 are considered;
- (2) San Antonio basin, Escondido Creek Sub. 11 near Kenedy (lat. 28°51'39", long. 97°50'39", code 08187900); the drainage area is 22.80 km²; elevations vary from 92 to 139 m, with a mean slope of 2.9%; 19 rainfall–runoff observed events from 1962–1970 are analysed;
- (3) Trinity basin, North Creek near Jacksboro (lat. 33°16'57", long. 98°17'53", code 08042700); the drainage area is 58.96 km²; elevations vary from 312 to 440 m, with a mean slope of 5.2%; 52 rainfall–runoff observed events from 1959–1979 are considered; and
- (4) Brazos basin, North Elm Creek near Cameron (lat. 30°55'52", long. 97°01'13", code 08108200); the drainage area is 119.46 km²; basin elevations vary from 101 to 188 m, with a mean slope of 1.4%; 20 rainfall–runoff observed events from 1964–1972 are used.

The digital elevation model (DEM) and other ancillary data, such as soil use, for the four basins are available on the US Geological Survey National Map Seamless Server (<http://seamless.usgs.gov/>). These basins were analysed within the project “Synthesis of Rainfall and Runoff Data Used for Texas Department of Transportation Research” (Asquith *et al.* 2004), in which they are classified as “small ruralsheds

database”, being characterized by limited human activity.

Table 1 shows the main morphometric properties of the four selected watersheds, while the topography and drainage network are displayed in Fig. 1.

The whole set of rainfall–runoff events considered in this study is listed in the Appendix (Table A1).

TIME OF CONCENTRATION ESTIMATION PROCEDURES

The time of concentration is estimated using the following methodologies: (a) direct estimation based on the observed rainfall–runoff data set through computational definitions; (b) estimation through empirical formulas; and (c) estimation based on the NRCS formula that includes flow velocity. Method (a) could be assumed as a direct “measurement” of T_c , while (b) and (c) are indirect estimation methods.

Direct estimation based on the observed rainfall–runoff data set

The time of concentration can be estimated using definition (i) presented in the Introduction from a selected rainfall–runoff observed flood event.

The procedure is apparently simple to implement; nevertheless, the estimation of the direct runoff, the inflection point and the rainfall excess are generally very challenging. Moreover, the subjectivity in their identification can limit the practical application and reliability of the procedure.

A recursive filter may be used to isolate the direct runoff (Lyne and Hollick 1979, Nathan and McMahon 1990), that is:

$$Q_d(t) = \beta Q_d(t-1) + \frac{1+\beta}{2} [Q(t) - Q(t-1)] \quad (1)$$

Table 1 Main morphometric properties of the four selected watersheds.

	Cow Bayou	Escondido Creek	North Creek	North Elm Creek
Area (km ²)	13.08	22.8	58.96	119.46
Min elevation (m)	180.3	92.1	312.4	100.9
Max elevation (m)	266.3	138.4	439.7	188.0
Mean elevation (m)	223.5	114.6	373.6	144.5
Max slope (%)	39.1	24.3	43.9	21.7
Mean slope (%)	5.9	2.9	5.2	1.4
Main channel length (km)	7.09	8.00	18.02	33.34
Max. divide–outlet distance (km)	7.35	8.63	18.47	35.55
Max order (-)	4	4	6	3

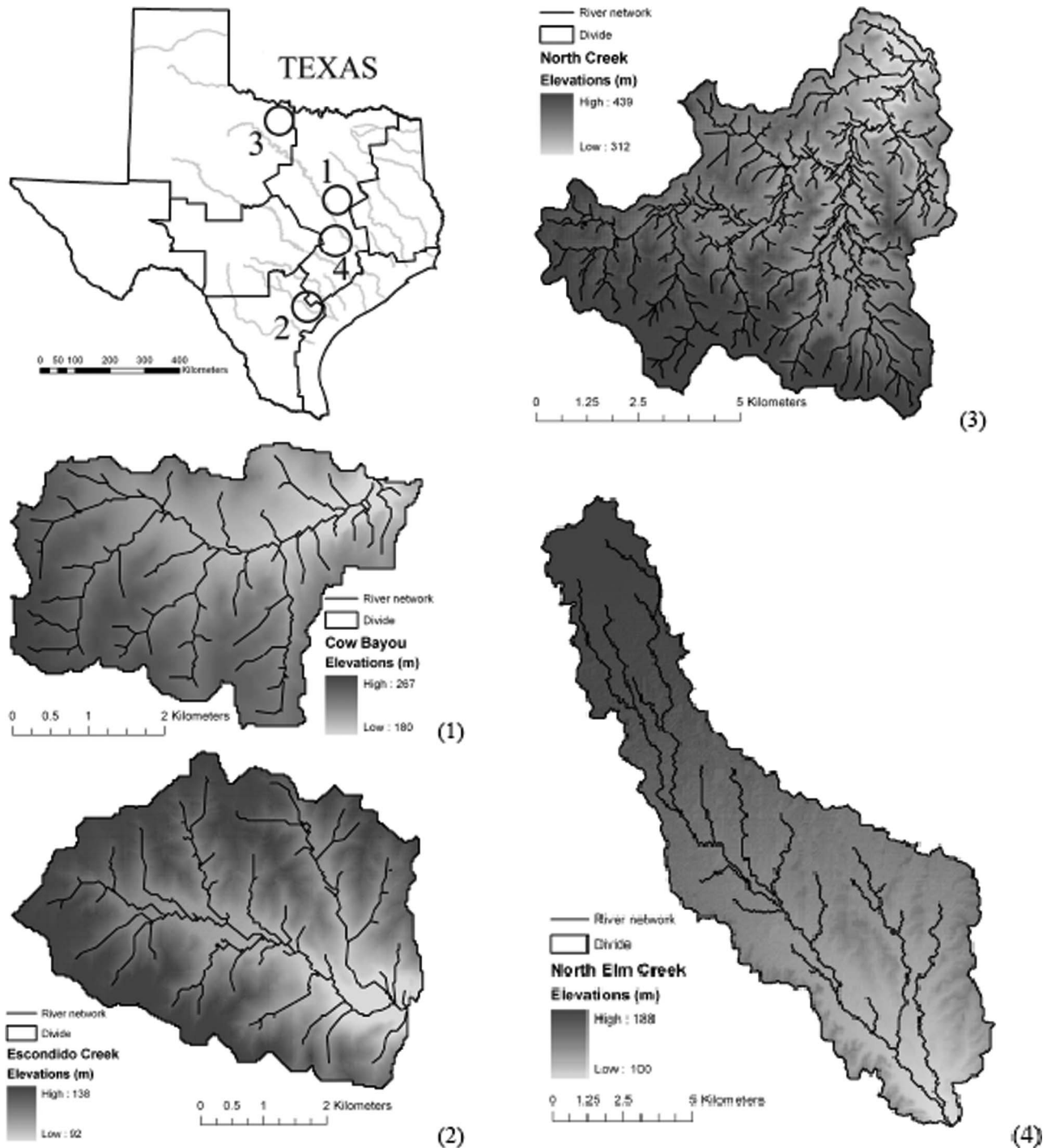


Fig. 1 Location and the topography and drainage networks of the four selected watersheds.

Here, $Q(t)$ and $Q(t-1)$ are the gross runoff at time t and $t-1$, $Q_d(t)$ and $Q_d(t-1)$ are the direct runoff at time t and $t-1$ (when $t=0$ the runoff is equal to zero), and β is the recursive filter parameter. The parameter values used in this analysis are given in the Appendix and their selection guarantees that the resulting inflection point is in line with the gross runoff recession curve. The recursive filter

is used three times (forward-backward-forward) to minimize phase distortion effects on the peak values (Nathan and McMahon 1990, Serinaldi and Grimaldi 2011).

The rainfall excess is obtained by using the widely used Soil Conservation Service curve number (SCS-CN) method (USDA 1986, Chow *et al.* 1988) where the CN parameter is selected so that the total

excess rainfall is equal to the total direct runoff volume.

Estimation using empirical formulas

Selected formulas find ample application in the technical literature due to the limited amount of information they require in estimating the time of concentration at basin scale in ungauged conditions. Here, such formulas are reported:

- Johnstone and Cross (1949):

$$T_c = 300 \sqrt{\frac{L}{S}} \quad (2)$$

where T_c is the time of concentration (min), L the maximum distance (miles) between the watershed divide and the outlet and S the basin slope (ft/mi). This formula is calibrated for drainage areas between 64 and 4200 km²;

- Department of Public Works (1995):

$$T_c = 60 (11.9L^3/H)^{0.385} \quad (3)$$

where T_c is the time of concentration (min), L the maximum distance (mi) between the watershed divide and the outlet and H the maximum elevation difference (ft) between the watershed divide and the outlet. This formula was originally developed for small mountain basins in California;

- NRCS (1997):

$$T_c = 0.0526 [(1000/CN) - 9]^{0.7} L^{0.8} S^{-0.5} \quad (4)$$

where T_c is the time of concentration (min), L the maximum distance (ft) between the watershed divide and the outlet, CN the curve number basin parameter (-), and S the basin mean slope (%). This formula was developed for small rural basins;

- Giandotti (1934):

$$T_c = \frac{4\sqrt{A} + 1.5L}{0.8\sqrt{H}} \quad (5)$$

where T_c is the time of concentration (h), A the watershed area (km²), L the length of the main channel (km) and H the difference between the mean basin elevation and the outlet elevation (m). This formula was calibrated on 12 basins with drainage areas between 170 and 70 000 km². It is

not specified how many and which flood events were used in the calibration procedure;

- Kirpich (1940):

$$T_c = 0.0078L^{0.77}S^{-0.385} \quad (6)$$

where T_c is the time of concentration (min), L the length of the main channel (ft) and S the mean basin slope (-). This formula was developed for small basins in Tennessee and Pennsylvania, USA; and

- Viparelli (1961, 1963):

$$T_c = L / (60V) \quad (7)$$

where T_c is the time of concentration (min), L the maximum distance (m) between the watershed divide and the outlet and V the mean flow channel velocity (m/s) with suggested values between 1 and 1.5 m/s (in the four case studies flow channel velocity is fixed to 1.5 m/s).

Estimation using NRCS flow velocity method

This method is based on the theoretical definition (i) mentioned in the Introduction. The flow velocities in the longest path to the outlet, required by the method, are quantified from the DEM watershed drainage network. A specific DEM cell can be classified either as “hillslope” or “channel” (see Grimaldi *et al.* 2010 and references therein). The flow velocity in these two classes is respectively computed using the following formulas (NRCS 1972, 1986):

$$v = a\sqrt{S} \quad (8)$$

$$v = \frac{R^{\frac{2}{3}}}{n} \sqrt{S} \quad (9)$$

where v is the flow velocity (m/s) in the single DEM cell, S is the cell slope (-), a is a coefficient (m/s) related to the soil use (Haan *et al.* 1994, McCuen 1998), n is the Manning roughness coefficient (s/m^{1/3}) and R the hydraulic radius (m). For $S > 0.04$, the cell slope in equation (8) is adjusted to:

$$S' = 0.05247 + 0.06363S - 0.182 \cdot e^{-62.38S} \quad (10)$$

in order to avoid flow velocity overestimation.

The use of equation (9) is the main cause of uncertainties for this method. Indeed, several geometrical hypotheses are required to define the hydraulic

radius of the channel section. It is not perfectly clear whether the analyst has to implement the method cell by cell or by discretizing the whole drainage network in segments and then applying an average geometrical hypothesis on the channel section. After selecting a section shape, either the water depth or the velocity should be defined in each cell or segment. This step can also induce relevant uncertainties. In conclusion, the time of concentration is estimated from the velocities cell by cell in the longest flow path.

While this approach is considered to be the most physically sound, it still suffers from a few major drawbacks, that is:

- the geometrical hypotheses underlying the use of Manning's formula (9) have a prominent influence on the final estimation. Fang *et al.* (2007) report an interesting application of this method that is particularly relevant to this specific challenge. The NRCS velocity method is used by three distinct research groups to estimate the time of concentration of 96 watersheds in Texas. These teams independently estimated the geometrical input parameters for T_c , including geometrical section and water depth and flow velocity. As a result, the average relative difference among the estimations of the time of concentration by them was 50% with peaks of 90% (Fang *et al.* 2007);
- the automatic extraction of the drainage network from DEM may significantly affect the time of concentration estimation process. A major problem is the blue line definition that is not univocally defined in literature (Nardi *et al.* 2008). Further uncertainties are also due to DEM resolution, indeed Pavlovic and Moglen (2008) show that higher time of concentration are obtained for high resolution DEMs; and
- the selection of the hydraulically longest path from the most distant point in the watershed to the outlet. Within this approach the estimation of basin scale parameters, such as T_c , is restrained to single cell identification.

Here, the NRCS flow velocity method is applied to the aforementioned case studies by adopting the following implementation:

- DEM pre-processing using the PEM4PIT procedure (Grimaldi *et al.* 2004 and 2007, Santini *et al.* 2009);

- drainage network identification using the procedure described in Grimaldi *et al.* (2010). Therein, the estimation of the flow directions is performed by using the single flow D8-LTD algorithm and the stream network automatic extraction is obtained via the curvature-based scheme in conjunction with the automated constant drop analysis algorithm for the identification of the channel initiation threshold;
- flow velocity estimation in hillslope cells using formulas (8) and (10); a range is fixed for minimum and maximum velocity values: 0.02–2 m/s;
- discharge estimation for a specific return period in each channel cell using the regression formulas described in Asquith and Slade (1997); such formulas involve only the contributing area and return period and they are calibrated for the case studies undertaken in this work; and
- velocity estimation for a specific return period in each channel cell assuming for each one the following attributes: a trapezoidal shape channel section with lateral slope equal to 2:1 (Fang *et al.* 2007: USGS and LU methods), a roughness coefficient equal to 0.04 (Fang *et al.* 2007: USGS method), and a channel width equal to 8.6 m (Fang *et al.* 2007, p. 316: Table 1). As a result, the channel average velocity is determined.

In the following, changes in T_c are ascertained as the DEM resolution, the most distant point of the flow path selection, and the return period related to the discharge and channel velocity, are varied.

RESULTS AND COMMENTS

Figure 2 shows T_c values estimated from the observed rainfall–runoff flood events data with the direct estimation procedure. For the Cow Bayou and North Creek basins, a slight asymptotic behaviour is visible. Increasing the discharge, similar T_c values are obtained. In practical terms, a reasonable T_c value for extreme hydrological analyses and for the calibration of empirical formulas should be the convergence value. These values correspond to approximately 1 and 3 hours for Cow Bayou and North Creek, respectively.

The remarkable data scatter in Fig. 2 demonstrates the inherent variability in T_c . Therefore, its estimation can vary up to 500% if only a partial event set is available.

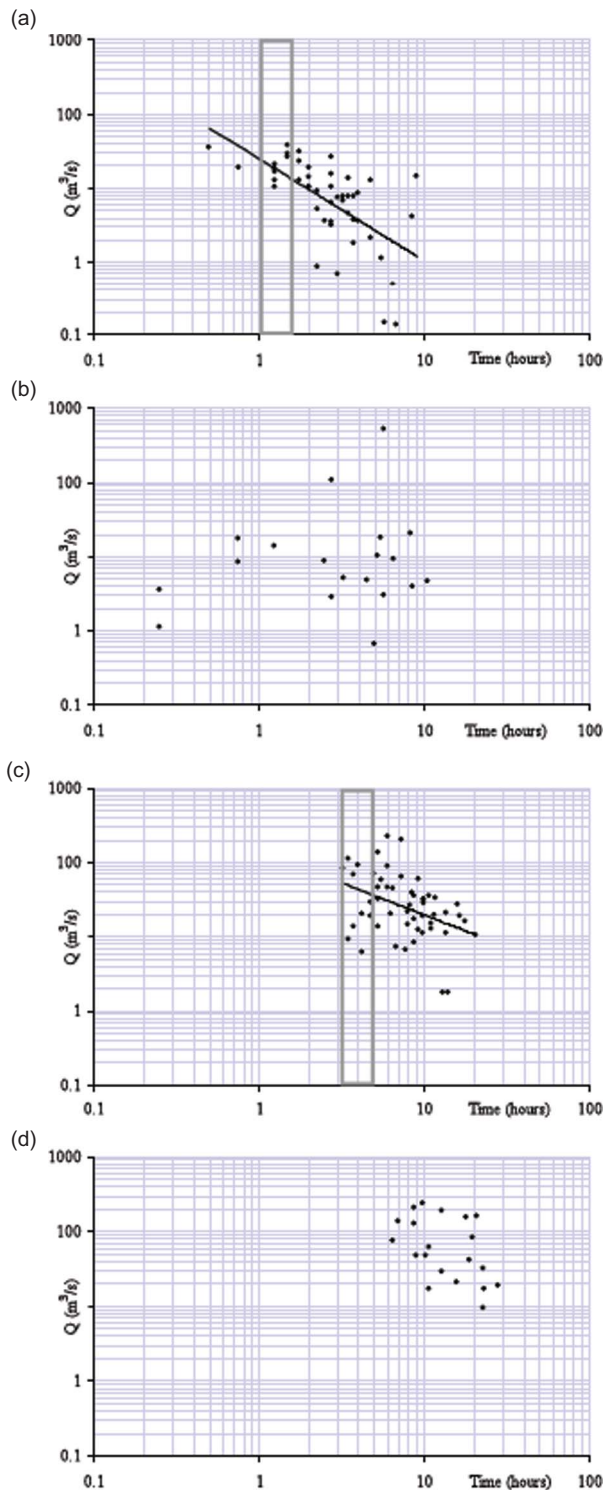


Fig. 2 Values of T_c estimated from the rainfall–runoff flood events observed in the four selected basins: (a) Cow Bayou, (b) Escondido Creek, (c) North Creek, and (d) North Elm Creek. In (a) and (c), a trend line is superimposed along with a tentative range for T_c identification.

Table 2 reports T_c values estimated using the empirical formulas (2)–(7). Consistently with the evidence in Fig. 2, the mere application of such empirical formulas yields prediction within a 500% range.

Table 3 lists the NRCS method results obtained by varying the DEM resolution from 5 to 30 m and using a 10-year return period discharge. The four DEMs are built using a cubic resampling starting from the original resolution of 30 m. As expected, increasing the resolution, that is, reducing the cell dimension, the flow path length and T_c values increase. These differences are moderate when compared to those obtained by varying other factors.

Table 4 displays the results of the NRCS method obtained by varying the return period discharge from 2 to 100 years using the original 30 m DEM resolution. An appreciable decrease of T_c is observed as the return period is increased in the studied range. Since T_c is largely used in extreme value analyses, results in Table 4 encourage the use of higher return period to avoid potential overestimation.

Table 5 reports NCRS method results obtained by varying the “most hydraulically distant” source point used for identifying the flow path. The first 10, 20, and 50 points are identified in terms of the hydraulic distance from the outlet. The source points share a similar flow path length to the outlet; nevertheless, the difference in slope distribution may be responsible for considerably different values of T_c .

Figure 3 displays aggregated data from Tables 1–5 in box plots to better highlight the dramatic variability in the T_c estimation; such variability is amplified by watershed dimension and can reach up to 500%.

CONCLUSIONS

In this paper, inherent uncertainties in the estimation of the time of concentration are demonstrated through an empirical analysis. A direct procedure on observed rainfall–runoff flood events, several empirical formulas, and the NRCS flow velocity method are implemented on four selected watersheds.

Obtained results are in line with the diffuse recognition of T_c variability. Nevertheless, they are somehow surprising, as they demonstrate dramatic variations in T_c estimation that can be as high as 500%. Furthermore, upper and lower bounds of T_c can not be *a priori* predicted and this variability is indeed due to discrepancies in the definitions and estimation procedures. Finally, a significant, and less expected, T_c variability as a function of the return periods and peak flows has been observed in the case study results. These findings support and motivate the title of this paper. Indeed, it is a paradox that advanced hydraulic models, such as 2-D flood propagation models for hydraulic risk mapping based on

Table 2 Values of T_c estimated with the empirical formulas (2)–(7) on the four selected basins.

Basin	Empirical formulas T_c (min)					
	Johnstone and Cross	California Culvert Practice	NRCS – SCS	Giandotti	Kirpich	Viparelli
Cow Bayou	119	98	223	286	55	82
Escondido Creek	183	144	380	492	81	96
North Creek	201	248	536	554	117	205
North Elm Creek	544	601	1457	1066	324	395

Table 3 Values of T_c estimated with the NRCS method varying the DEM resolution from 5 to 30 m.

Basin	DEM resolution (m)	T_c (min)
Cow Bayou	5	153
	10	89
	20	82
	30	98
Escondido Creek	5	241
	10	222
	20	229
	30	244
North Creek	5	315
	10	324
	20	270
	30	253
North Elm Creek	5	1276
	10	1047
	20	927
	30	946

Table 4 Values of T_c estimated with the NRCS method varying the return period discharge (Tr) from 2 to 100 years.

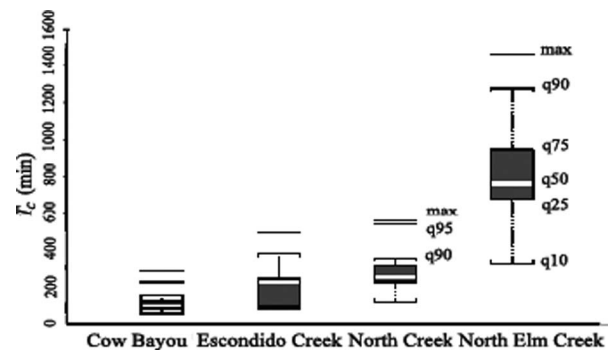
Basin	T_c (min)					
	Tr 2	Tr 5	Tr 10	Tr 25	Tr 50	Tr 100
Cow Bayou	121	103	98	93	91	88
Escondido Creek	219	251	244	238	235	232
North Creek	353	282	253	231	222	214
North Elm Creek	936	838	801	770	756	754

very expensive topographic and remote sensing data, are actually limited by design hydrographs based on anachronistic parameters, such as T_c .

Building upon the critical assessment of available definition and estimation procedure, possible amelioration and research values can be identified. For instance, presented results suggest that the estimation of NRCS method can be improved through proper selection of the most hydraulically-distant point and the channel discharge values. It is also expected that

Table 5 Values of T_c estimated with the NRCS method varying the number of most hydraulically distant source points (N).

Basin	T_c (min)								
	N 10			N 20			N 50		
	min	mean	max	min	mean	max	min	mean	max
Cow Bayou	59	61	63	58	60	63	51	63	93
Escondido Creek	89	91	94	87	90	94	81	87	94
North Creek	193	216	230	192	213	230	186	211	253
North Elm Creek	640	657	682	638	657	682	628	655	682

**Fig. 3** Box plots of all estimated T_c values for the four selected watersheds.

technological advancement in the field of direct flow measurements in hillslope and small channels will fundamentally contribute to alleviate this paradox.

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REFERENCES

- Aronica, G.T. and Candela, A., 2007. Derivation of flood frequency curves in poorly gauged Mediterranean catchments using a simple stochastic hydrological rainfall-runoff model. *Journal of Hydrology*, 347, 132–142.
- Asquith, W.H. and Slade, R. Jr, 1997. Regional equations for estimation of peak-stream flow frequency for natural basins in Texas. *US Geological Survey Water-Resources Investigations Report 96-4307* (<http://pubs.usgs.gov/wri/wri964307/>).
- Asquith, W.H., Thompson, D.B., Cleveland, T.G. and Fang, X., 2004. Synthesis of rainfall and runoff data used for Texas. Austin, TX: Department of Transportation Research Projects 0-4193 and 0-4194, *US Geological Survey Open File Report 2004-1035*.
- Calkins, D. and Dunne, T., 1970. A salt tracing method for measuring channel velocities in small mountain streams. *Journal of Hydrology*, 11 (4), 379–392.
- Chow, V.T., Maidment, D.R. and Mays, L.W., 1988. *Applied hydrology*. New York: McGraw-Hill.
- Department of Public Works, 1995. *California culvert practice, second edition*. Sacramento, CA: DPW, Division of Highways.
- Dooge, J.C.I., 1973. Linear theory of hydrologic systems. US Department of Agriculture, *Technical Bulletin 1468*.
- Fang, X., Thompson, D.B., Cleveland, T.G. and Pradhan, P., 2007. Variations of time of concentration estimates using NRCS velocity method. *Journal of Irrigation and Drainage Engineering*, 133 (4), 314–322.
- Fang, X., Thompson, D.B., Cleveland, T.G., Pradhan, P. and Malla, R., 2008. Time of concentration estimated using watershed parameters determined by automated and manual methods. *Journal of Irrigation and Drainage Engineering*, 134 (2), 202–211.
- FEMA, 2010. Hydrologic models meeting the minimum requirement of NFIP. List of current nationally accepted hydrologic models by FEMA available on http://www.fema.gov/plan/prevent/fhm/en_hydro.shtm.
- Folmar, N.D., Miller, A.C. and Woodward, D.E., 2007. History and development of the NRCS lag time equation. *Journal of the American Water Resources Association*, 43 (3), 829–838.
- Giandotti, M., 1934. Previsione delle piene e delle magre dei corsi d'acqua. *Istituto Poligrafico dello Stato*, 8, 107–117.
- Grimaldi, S., Teles, V. and Bras, R.L., 2004. Sensitivity of a physically-based method of terrain interpolation to variance of input data and to the constraint of an enforced surface drainage direction, *Earth Surface Processes and Landforms*, 29(5), 587–597.
- Grimaldi, S., Nardi, F., Di Benedetto, F., Istanbuloglu, E. and Bras, R.L., 2007. A physically-based method for removing pits in digital elevation models. *Advances in Water Resources*, 30 (10), 2151–2158.
- Grimaldi, S., Petroselli, A., Nardi, F. and Alonso, G., 2010. Flow time estimation with variable hillslope velocity in ungauged basins. *Advances in Water Resources*, 33 (10), 1216–1223.
- Haan, C.T., Barfield, B.J. and Hays, J.C., 1994. *Design hydrology and sedimentology for small catchments*. New York: Academic Press.
- Holecek, J. and Vogel, J., 1965. Measurement of time of travel in the Volynka Experimental Basin. In: *Representative and experimental areas*, IAHS Symposium of Budapest, 28 September–1 October 1965. Wallingford, UK: IAHS Press, IAHS Publ. 61, 299–310 [Available at: <http://iahs.info/redbooks/066.htm>].
- Kirpich, Z.P., 1940. Time of concentration of small agricultural watersheds. *Civil Engineering*, 10 (6), 362.
- Koutroulis, A.G. and Tsanis, I.K., 2010. A method for estimating flash flood peak discharge in a poorly gauged basin: Case study for the 13–14 January 1994 flood, Giofiros basin, Crete Greece. *Journal of Hydrology*, 385, 150–164.
- Johnstone, D. and Cross, W.P., 1949. *Elements of applied hydrology*. New York: Ronald Press.
- Lyne, V. and Hollick, M., 1979. Stochastic time-variable rainfall-runoff modelling. *Institute of Engineers Australia National Conference*. Publ. 79/10, 89–93.
- McCuen, R.H., 1998. *Hydrologic design and analysis*. New Jersey: Prentice Hall.
- McCuen, R.H., 2009. Uncertainty analyses of watershed time parameters. *Journal of Hydrologic Engineering*, 14 (5), 490–498.
- Nardi, F., Grimaldi, S., Santini, M., Petroselli, A. and Ubertini, L., 2008. Hydrogeomorphic properties of simulated drainage patterns using DEMs: the flat area issue. *Hydrological Sciences Journal*, 53 (6), 1176–1193.
- Nathan, R.J. and McMahon, T.A., 1990. Evaluation of automated techniques for baseflow and recession analysis. *Water Resources Research*, 26 (7), 1465–1473.
- Neitsch, S.L. et al., 2005. SWAT—Soil and Water Assessment tool theoretical documentation, available at <http://swatmodel.tamu.edu/documentation>.
- NRCS (National Research Conservation Service), 1972. Hydrology. In: *National engineering handbook*, Sec. 4. Washington, DC: US Department of Agriculture.
- NRCS (National Research Conservation Service), 1986. Urban hydrology for small watersheds. Washington, DC: US Department of Agriculture, Technical Release no. 55.
- NRCS (National Research Conservation Service), 1997. *Ponds—Planning, design, construction*. Washington DC: US Natural Resources Conservation Service, Agriculture Handbook no. 590.
- PAI, 2006. *Piano Stralcio di Assetto Idrogeologico*. Autorità di Bacino del Fiume Tevere. Available in Italian at http://www.abtevere.it/attivita/piani_approvati/PAI_NEW/pai.htm.
- Pavlovic, S.B. and Moglen, G.E., 2008. Discretization issues in travel time calculation. *Journal of Hydrologic Engineering*, 13 (2), 71–79.
- Pilgrim, D.H., 1975. Travel times and nonlinearity of flood runoff from tracer measurements on a small watershed. *Water Resources Research*, 12, 487–496.
- Santini, M., Grimaldi, S., Rulli, M.C., Petroselli, A. and Nardi, F., 2009. Pre-Processing algorithms and landslide modelling on remotely sensed DEMs. *Geomorphology*, 113(1-2), 110–125.
- Seybert, T.A., 2006. *Stormwater management for land development: methods and calculations for quality control*. New York: John Wiley and Sons.
- Serinaldi, F. and Grimaldi, S., 2011. Synthetic design hydrographs based on distribution functions with finite support. *Journal of Hydrologic Engineering*, 16 (5), 434–446, doi: 10.1061/(ASCE)HE.1943-5584.0000339.
- USACE (US Army Corps of Engineers), 2001. *HEC-HMS hydrologic modeling system, user's manual, version 2.2.1*. Vicksburg, Mississippi: US Army Corps of Engineers.
- USDA-SCS (US Department of Agriculture – Soil Conservation Service), 1986. *Urban hydrology for small watersheds*. Washington, DC: US Department of Agriculture, Soil Conservation Service, Technical release 55.
- Viparelli, C., 1961. *Ricostruzione dell'idrogramma di piena*. Napoli: Istituto di Idraulica dell'Università di Palermo, Stab. Tip. Genovese (in Italian).
- Viparelli, C., 1963. *Ricostruzione dell'idrogramma di piena*. *L'Energia Elettrica*, 6, 421–428 (in Italian).
- Wong, T.S.W., 2005. Assessment of time of concentration formulas for overland flow. *Journal of Irrigation and Drainage Engineering*, 131 (4), 383–387.

APPENDIX

Table A1 Data set of rainfall–runoff observations for the four selected watersheds (CGRF: cumulated gross rainfall; CGRU: cumulated gross runoff; β : recursive filter parameter of formula (1); CN: curve number parameter; CNRU: cumulated net runoff; gross $Q_{\max}$: maximum gross runoff; net $Q_{\max}$: maximum direct runoff).

Event (-)	Date (mm/dd/yy)	CGRF (mm)	CGRU (mm)	β (-)	CN (-)	CNRU (mm)	Gross $Q_{\max}$ (m ³ /s)	Net $Q_{\max}$ (m ³ /s)
Cow Bayou:								
1	10/03/1959	157.0	38.4	0.998	28.9	31.5	39.1	37.9
2	12/15/1959	64.0	15.9	0.985	37.2	8.3	29.2	26.3
3	12/06/1960	144.8	69.7	0.999	40.5	40.5	15.1	14.3
4	02/05/1961	66.6	24.4	0.996	47.3	12.7	11.6	10.4
5	06/08/1961	65.3	6.5	0.989	23.2	4.7	17.7	16.6
6	11/21/1961	37.3	2.9	0.992	20.5	1.4	3.3	3.1
7	04/27/1962	32.0	0.4	0.993	5.5	0.2	0.2	0.1
8	05/28/1962	72.4	0.9	0.998	3.6	0.8	0.5	0.5
9	06/01/1962	29.2	1.1	0.995	21.5	0.9	3.7	3.6
10	06/30/1962	43.4	4.1	0.994	30.9	3.1	8.0	7.5
11	03/18/1964	39.6	2.0	0.997	20.3	1.5	2.2	2.1
12	04/26/1964	16.3	0.6	0.975	16.0	0.2	0.8	0.7
13	06/15/1964	122.9	3.1	0.998	4.3	2.6	4.1	4.0
14	01/21/1965	71.4	5.1	0.998	18.1	4.2	3.8	3.6
15	03/29/1965	109.5	26.5	0.995	28.6	16.1	30.8	28.9
16	05/09/1965	23.4	0.2	0.992	5.9	0.1	0.2	0.1
17	05/10/1965	64.0	11.7	0.991	33.7	7.3	22.4	20.9
18	02/08/1966	60.7	18.9	0.988	42.4	9.1	37.9	35.8
19	04/24/1966	50.0	9.3	0.991	36.9	5.2	19.6	18.6
20	04/24/1966	35.3	10.9	0.995	63.5	6.9	9.0	8.4
21	04/25/1966	13.5	8.1	0.990	83.5	2.8	5.2	4.5
22	05/20/1966	41.7	12.9	0.992	55.0	6.9	15.0	13.6
23	05/21/1966	18.3	5.4	0.982	70.1	2.6	13.6	12.6
24	09/17/1967	19.3	0.4	0.970	10.5	0.2	1.0	0.9
25	11/09/1967	71.6	91.4	0.999	22.3	5.4	3.8	3.7
26	06/23/1968	53.9	2.1	0.999	12.8	1.6	1.2	1.1
27	07/02/1968	24.6	1.0	0.989	23.5	0.7	2.0	1.8
28	07/08/1968	58.9	10.0	0.996	40.9	8.2	23.7	23.1
29	03/22/1969	34.0	6.9	0.995	53.3	4.5	13.1	12.6
30	04/12/1969	55.4	13.9	0.993	44.4	8.2	8.9	7.6
31	04/17/1969	22.9	4.9	0.961	48.9	1.8	6.4	5.2
32	05/05/1969	26.9	6.6	0.992	60.6	3.8	9.7	9.1
33	12/05/1969	58.4	7.0	0.997	28.7	5.0	7.0	6.7
34	02/23/1970	46.0	8.3	0.995	39.6	4.9	8.4	7.8
35	03/06/1970	30.7	9.9	0.990	62.9	5.2	11.7	10.4
36	03/07/1970	16.3	9.3	0.992	87.5	5.1	8.8	7.8
37	03/16/1970	22.4	7.9	0.987	66.4	3.3	15.0	14.0
38	07/25/1971	84.8	7.7	0.989	16.7	5.3	13.8	12.5
39	11/17/1971	90.9	8.9	0.988	16.1	5.8	20.2	18.7
40	12/09/1971	34.3	8.4	0.993	48.8	3.9	4.2	3.5
41	03/24/1973	55.9	16.1	0.988	44.7	8.5	20.5	18.0
42	05/25/1973	64.3	6.2	0.975	18.1	3.4	12.2	10.3
43	06/03/1973	67.6	18.9	0.991	45.6	12.3	33.1	30.8
44	10/11/1973	70.9	9.1	0.996	27.7	6.8	16.0	15.4
45	10/31/1974	90.9	24.2	0.997	42.3	18.9	27.4	26.3
46	11/23/1974	36.6	7.3	0.991	34.5	2.6	7.0	6.4
47	02/01/1975	67.6	21.4	0.996	44.9	12.0	11.7	10.3
Escondido Creek:								
1	12/02/1962	46.7	4.2	0.989	19.5	2.0	9.1	8.2
2	12/20/1962	42.7	0.7	0.973	2.7	0.2	1.4	1.1
3	06/25/1963	48.5	5.8	0.990	30.6	3.8	18.5	17.0

(continued).

Table A1 (Continued).

Event (-)	Date (mm/dd/yy)	CGRF (mm)	CGRU (mm)	β (-)	CN (-)	CNRU (mm)	Gross $Q_{\max}$ (m ³ /s)	Net $Q_{\max}$ (m ³ /s)
4	11/27/1963	60.5	1.3	0.982	4.6	0.7	3.3	2.8
5	12/12/1963	56.1	0.8	0.995	4.4	0.6	0.7	0.7
6	03/18/1964	45.0	2.1	0.996	16.5	1.5	4.9	4.7
7	08/08/1964	81.8	2.6	0.996	7.5	2.1	5.3	5.0
8	02/04/1965	74.2	7.5	0.998	22.8	5.9	9.1	8.5
9	02/16/1965	56.4	11.2	0.995	39.7	7.2	12.0	10.3
10	05/11/1965	85.1	6.7	0.989	13.1	4.1	15.5	13.7
11	05/19/1965	68.6	35.1	0.989	63.9	22.2	115.6	105.8
12	09/17/1966	64.3	3.0	0.997	12.7	2.3	9.5	9.2
13	09/19/1967	489.2	425.8	0.998	42.2	285.9	530.5	510.3
14	05/07/1968	38.9	0.3	0.970	2.6	0.2	3.7	3.5
15	05/11/1968	119.4	30.2	0.997	30.8	20.7	22.2	20.4
16	03/14/1969	51.6	3.2	0.996	16.3	2.0	4.8	4.6
17	05/04/1969	43.2	7.1	0.990	27.9	2.7	5.2	3.8
18	05/28/1970	21.84	2.5	0.994	43.5	1.4	3.3	2.9
19	05/31/1970	40.39	11.4	0.988	50.2	5.6	20.9	18.1
North Creek:								
1	10/03/1959	214.4	72.1	0.998	29.8	56.4	214.3	197.2
2	06/09/1962	108.7	58.9	0.996	61.5	44.1	234.8	220.9
3	07/27/1962	76.7	18.1	0.997	42.0	13.8	73.5	69.4
4	09/05/1962	43.2	11.5	0.999	63.2	9.8	29.2	25.6
5	09/07/1962	34.0	12.4	0.996	71.4	8.5	31.7	27.9
6	11/26/1962	53.6	11.4	0.998	46.8	8.4	23.6	21.4
7	04/26/1963	74.7	9.3	0.998	26.8	7.3	47.1	45.1
8	04/28/1963	28.4	13.7	0.995	78.1	8.1	98.5	88.5
9	05/27/1963	29.5	3.4	0.998	46.0	2.6	20.0	19.6
10	11/19/1963	73.9	4.1	0.997	13.8	3.3	29.7	28.8
11	05/29/1964	114.8	22.8	0.997	27.6	16.8	48.6	45.8
12	11/16/1964	48.0	3.0	0.996	19.9	2.2	14.5	13.5
13	11/17/1964	53.3	10.6	0.996	41.9	7.0	35.3	31.8
14	11/18/1964	30.2	11.3	0.995	74.1	7.7	48.2	44.0
15	05/12/1965	31.0	10.7	0.999	76.9	9.0	39.6	38.5
16	05/13/1965	17.0	4.2	0.997	75.0	2.9	13.6	12.7
17	05/18/1965	25.1	4.9	0.998	60.9	3.4	16.3	15.7
18	09/18/1965	107.7	17.8	0.998	28.0	15.2	67.2	64.1
19	05/29/1967	63.5	13.8	0.998	44.3	10.5	34.3	32.0
20	05/30/1967	49.5	12.5	0.999	57.1	10.2	34.4	33.0
21	07/05/1967	23.6	1.9	0.994	42.2	1.5	21.7	20.9
22	09/21/1967	26.9	3.3	0.998	47.9	2.4	12.9	12.3
23	01/21/1968	12.7	2.2	0.994	62.5	1.0	2.2	1.8
24	03/11/1968	49.0	9.3	0.999	47.7	7.3	10.7	10.5
25	06/05/1968	38.1	4.4	0.996	36.2	3.0	19.6	18.4
26	07/08/1968	58.4	2.3	0.993	9.3	1.3	14.3	13.5
27	07/18/1968	36.8	2.7	0.998	28.9	2.1	11.8	11.2
28	03/07/1969	14.7	1.5	0.996	55.5	1.0	8.5	8.1
29	03/14/1969	51.3	8.2	0.998	39.3	5.9	12.3	11.1
30	03/23/1969	40.6	8.3	0.987	36.7	3.5	23.5	18.6
31	05/04/1969	70.6	24.2	0.997	55.3	18.1	88.7	82.5
32	05/06/1969	45.2	21.8	0.999	78.4	17.7	60.5	58.6
33	12/28/1969	65.3	7.6	0.999	27.8	5.9	16.8	15.0
34	03/06/1970	23.1	1.0	0.994	18.0	0.4	2.0	1.8
35	04/25/1970	40.1	10.3	0.999	63.1	8.5	35.8	34.8
36	04/30/1970	40.6	23.8	0.994	78.1	14.7	99.7	89.8
37	09/22/1971	74.4	6.3	0.996	17.3	4.3	19.3	18.5
38	04/27/1972	34.5	1.2	0.997	15.6	0.8	6.8	6.5
39	05/11/1972	89.7	28.0	0.993	34.0	13.8	118.1	109.7
40	07/28/1973	88.6	2.6	0.999	7.3	2.4	14.5	14.3
41	07/30/1973	40.4	3.3	0.998	31.8	2.8	17.5	17.1

(continued).

Table A1 (Continued).

Event (-)	Date (mm/dd/yy)	CGRF (mm)	CGRU (mm)	β (-)	CN (-)	CNRU (mm)	Gross $Q_{\max}$ (m ³ /s)	Net $Q_{\max}$ (m ³ /s)
42	08/29/1974	71.6	4.3	0.998	16.1	3.7	20.5	19.8
43	10/30/1974	69.9	25.9	0.994	41.1	11.2	58.8	57.7
44	05/02/1975	51.3	12.7	0.998	50.5	8.8	29.0	27.0
45	08/26/1975	50.8	4.1	0.998	26.8	3.5	20.3	19.7
46	04/19/1976	34.8	1.7	0.998	23.8	1.4	9.3	9.1
47	09/19/1976	121.2	16.0	0.997	15.0	9.4	37.6	34.6
48	03/26/1977	87.4	21.0	0.993	21.0	7.3	36.3	30.5
49	05/23/1977	50.5	25.8	0.998	72.8	17.5	139.8	136.1
50	04/09/1978	120.7	16.9	0.998	15.8	9.9	70.7	68.2
51	08/05/1978	30.0	1.0	0.998	20.4	0.9	7.3	7.2
52	04/17/1979	53.6	1.9	0.997	11.0	1.4	6.5	6.1
North Elm Creek:								
1	09/24/1964	47.5	3.1	0.997	17.7	1.8	10.6	9.3
2	01/21/1965	106.17	25.2	0.997	30.9	16.8	244.6	232.3
3	04/25/1965	39.62	12.1	0.999	64.8	8.9	50.1	47.5
4	05/16/1965	95.25	99.2	0.995	74.3	49.6	257.6	209.0
5	05/17/1965	9.65	5.1	0.999	97.2	4.3	22.3	20.9
6	05/18/1965	7.62	4.7	0.998	96.9	3.7	19.0	16.8
7	05/28/1965	74.42	47.3	0.999	80.4	40.6	158.6	152.9
8	04/24/1966	102.62	93.7	0.999	82.9	68.0	167.7	157.8
9	07/08/1968	64.77	42.2	0.996	73.9	27.2	161.4	134.3
10	11/30/1968	34.54	15.1	0.998	80.1	12.2	45.1	41.5
11	04/11/1969	56.64	52.0	0.998	89.3	36.8	200.9	187.7
12	10/29/1969	98.55	23.4	0.994	30.6	14.4	146.8	125.7
13	02/06/1970	35.31	17.5	0.999	82.7	14.1	63.4	60.3
14	09/26/1970	36.32	8.3	0.999	61.1	6.7	20.2	18.6
15	10/22/1970	58.42	16.6	0.999	56.1	13.3	49.6	47.0
16	05/08/1971	61.98	33.2	0.998	72.7	24.5	96.4	82.4
17	05/27/1971	14.48	6.4	0.999	91.3	5.4	32.7	31.9
18	11/17/1971	81.03	19.6	0.999	45.3	17.0	78.2	74.8
19	04/29/1972	17.78	4.1	0.998	76.9	3.4	29.7	28.3
20	05/01/1972	27.18	5.7	0.999	67.8	5.0	17.2	16.7